

	MODE 1	MODE 2	MODE 3	MODE 4	UNITS
RANGE_UNITS	X	X	X		
TRANSMIT_DELAY_1	X	X	X		s
RECEIVE_DELAY_1	X	X	X		s
CORRECTION_RANGE	X	X	X		s
CORRECTION_DOPPLER		X	X	X	Hz
CORRECTIONS_APPLIED	X	X	X	X	
PATH	X	X	X	X	

MODE is always set to SEQUENTIAL, which indicates sequential paths from ground to spacecraft, and back to ground.

TRANSMIT_BAND and **RECEIVE_BAND** are set to S, X, or Ka, as appropriate.

TURANROUND_NUMERATOR and **TURNAROUND_DENOMINATOR** indicate the transponder turnaround ratio in use by the spacecraft transponder. Only standard turnaround ratios as defined in CCSDS 401.0-B-32 are supported. The turnaround ratio is defined by the spacecraft, but must be configured accurately on the ground to enable proper carrier phase and Doppler measurements.

INTEGRATION_INTERVAL defines the integration time for each radiometric measurement. Valid request values range from 10ms to 60s, although supported values by any given ranging system may vary. The user requests an integration time when ranging is enabled, and the nearest supported integration time is selected and reported in this element.

INTEGRATION_REF defines the relation of the measurement timetag to the integration interval used to produce the measurement. For all existing ranging systems supported by KSAT, the value will be END, indicating that the measurement is timestamped at the end of the integration interval.

RANGE_MODULUS (need for this is TBD).

RANGE_UNITS is set to s to indicate that range measurements are reported in seconds of round-trip delay.

TRANSMIT_DELAY_1 defines the nominal transmit delay of the signal from the calibration plane on the uplink test coupler to the reflector reference plane.

RECEIVE_DELAY_1 defines the nominal receive delay of the signal from the reflector reference plane to the calibration plane on the downlink test coupler.

CORRECTION_RANGE provides the range correction factor (in seconds) that is determined during pre-pass calibration. This correction includes both transmit and receive path calibration, and is used to adjust the ranging delay measurements prior to processing by flight software.

CORRECTION_DOPPLER provides the Doppler correction factor (in Hz) that is determined during pre-pass calibration.

CORRECTIONS_APPLIED is always set to NO, indicating that the radiometric measurements are not adjusted for any of the correction factors. The user should process the raw measurements as described in section 4 of this document to obtain actual range and range rate values suitable for flight dynamics.

PATH is always set to 1,2,1, indicating two-way, round-trip functionality. One-way operations, multiple receive sites (e.g. VLBI) and bistatic (separate uplink and downlink) modes are not supported.

Data Items

Data elements in the data subsection are listed in the following table.

	MODE 1	MODE 2	MODE 3	MODE 4	UNITS
RANGE	X	X	X		s
TRANSMIT_FREQ_1			X	X	Hz
RECEIVE_FREQ_1			X	X	Hz

The timestamp resolution for all data items shall be 1 us.

The **RANGE** keyword is used to report ranging measurements from the receiver. In spite of the name, these measurements are actually the round-trip time of the ranging signal, and corresponds to the T_m quantity as described in section TBD. Ranging measurements should be processed as described in this section to obtain the actual estimated range.

The **TRANSMIT_FREQ** and **RECEIVE_FREQ** are used to indicate the transmit and receive carrier frequencies, respectively. These are used to represent Doppler measurements, and correspond to the quantities F_{ut} and F_{dr} described in section TBD. The transmit frequency is reported once per file, unless it is adjusted for Doppler pre-steering.

Data Item Mapping

Several data items (including metadata) are mapped back to models described in other sections of this document. The mapping of each item to the appropriate modelling section, and the associated symbol, is shown in the following table.

DATA ITEM	LINK	SYMBOL
TRANSMIT_DELAY_1	TBD	Ttx
RECEIVE_DELAY_1	TBD	Trx
CORRECTION_RANGE	TBD	Tgc
CORRECTION_DOPPLER	TBD	
TURNAROUND_NUMERATOR	TBD	trn
TURNAROUND_DENOMINATOR	TBD	trd
RANGE	TBD	Tm
TRANSMIT_FREQ_1	TBD	Fut
RECEIVE_FREQ_1	TBD	Fdr